Typing the rare neurons of an unlabeled connectome by graph matching, guided by a transferred graph neural network

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October 2026

Abstract

New connectome reconstructions are typed by matching their neurons to an annotated reference. Most cell types are rare, with two to four neurons per brain, and methods that type neurons by their connections need labeled partners in the target. We type rare neurons without any target label in two stages. A graph neural network (GNN) that sees only anonymous wiring is trained on the reference and transferred to the target, where it selects candidate neurons; graph matching against the reference's rare-type neurons then assigns their types. In tests whose protocols were committed before any run on the held-out data, accuracy on rare-type neurons rose from 22 to 41 % on the sparse BANC brain and, with an improved candidate rule, from 40 to 61 % and from 72 to 82 % in the nerve cords of BANC and of the male CNS. Wiring complements morphology. In an analysis fixed in advance but run on BANC, whose labels we had already seen, combining wiring with unreviewed automatic morphology matches (NBLAST) beat these matches alone by 2.3 points on the brain (95 % CI 1.3–3.2), short of the 3 points we required, and by 23 points in the nerve cord, where the unreviewed matches are weak. BANC's reference types come from reviewed NBLAST matches, which favors NBLAST in this comparison.

1 Introduction

Electron-microscopy connectomes of the adult *Drosophila* central nervous system now exist for several individuals: the female FAFB/FlyWire brain (Dorkenwald et al., 2024; Schlegel et al., 2024), the hemibrain (Scheffer et al., 2020), the male ventral nerve cord MANC (Takemura et al., 2024; Marin et al., 2024), the male CNS (MCNS; Berg et al., 2026) and the female brain and nerve cord BANC (Bates et al., 2026). Their value rests on cell-type annotations. New reconstructions are typed by matching their neurons to an annotated reference, by morphology and by connectivity (Schlegel et al., 2024; Bates et al., 2026). Most types are small: of FlyWire's 8,772 cell types, about three quarters have two to four neurons. Connectivity-based typing compares each neuron's synaptic partners by type, as in the definition of optic-lobe types (Matsliah et al., 2024) and in NTAC (Schwartzman et al., 2026), and therefore needs labeled partners in the target.

We type these rare neurons in a target in which no neuron is labeled, using only its anonymous wiring diagram: the directed graph of neurons and synapse counts. A classifier trained on the reference has seen one or two examples of each rare type. But these types were defined by pairing

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neurons across hemispheres and brains, and pairing neurons is what graph matching does without labels (Pedigo et al., 2023). Graph matching needs to know which target neurons to pair, though. We therefore let a graph neural network (GNN), trained on the reference and transferred to the target without labels, decide which neurons to match, and graph matching decide their types (Figure 1a). Our contributions:

1. **Graph matching types rare neurons once a transferred GNN has chosen the candidates.** Within FlyWire, on the same candidates, matching types 92 % of the right-hemisphere neurons of 2–4-neuron types, against 80 % for the GNN; without any label, the two stages raise these types from 79 to 86 %. From FlyWire to the male CNS they rise from 54 to 82 % (Section 4.1).
2. **The gain holds on held-out connectomes, and candidate selection is the bottleneck.** In three pre-registered tests, on BANC's brain and from MANC to the nerve cords of BANC and MCNS, the two stages gained 10–21 points over the GNN alone (Section 4.2). Matching with perfectly chosen candidates would gain another 5–25 points (Section 4.3).
3. **Wiring complements morphology.** The two err on different neurons. Combined with unreviewed automatic morphology matches, wiring adds little where these matches are good (BANC brain, +2.3 points in a pre-specified analysis that missed its 3-point rule) and a lot where they are weak (BANC cord, +23 points). Morphology does not make the GNN redundant: without its candidates, the combination falls below NBLAST alone on the brain. BANC's reference types are partly NBLAST's own answers, which tilts the comparison toward morphology (Section 4.4).

We report all four pre-registered tests on held-out data and one pre-specified analysis on data whose labels we had seen, including what failed in them (Table 7): the first rule for choosing the GNN's preprocessing, a fallback for unmatched neurons, a threshold set too high, seeding the matching with morphology matches, and typing a partial volume.

2 Related work

Cell typing from connectomes. Fly connectomes have been annotated with thousands of types from morphology and connectivity (Scheffer et al., 2020; Schlegel et al., 2024; Matsliah et al., 2024; Marin et al., 2024). In the optic lobe, types were defined from each neuron's input and output vectors over types (Matsliah et al., 2024). NTAC (Schwartzman et al., 2026) automates this from labeled seeds (97.0 % on 434 FlyWire types in our reimplementation, Supplement S2) and has an unsupervised mode. Learned approaches mostly use morphology (Troidl et al., 2024) or combine it with connectivity (NeuNet; Liao et al., 2024). Across datasets, types are matched by NBLAST (Costa et al., 2016), connectivity alignment and combinations of both (Schlegel et al., 2024; Bates et al., 2026; Berg et al., 2026). These typing methods read partner labels or are evaluated within one labeled graph; we know of no evaluation of a trained typing model transferred to an unlabeled connectome.

Graph matching and label-free structure. Graph matching finds the correspondence between the nodes of two graphs that best aligns their edges; the relaxed quadratic assignment problem is solved efficiently by Frank–Wolfe iterations (FAQ; Vogelstein et al., 2015), and seeded graph matching fixes part of the correspondence in advance (Fishkind et al., 2019). Bisected graph matching, which also aligns the connections between hemispheres, pairs bilaterally homologous neurons of the larval connectome from connectivity alone (Pedigo et al., 2023). Adjacency spectral embedding with Gaussian mixtures recovers neuron groups in the larval mushroom body without labels (Priebe et al., 2017). These methods match or cluster neuron by neuron; we use

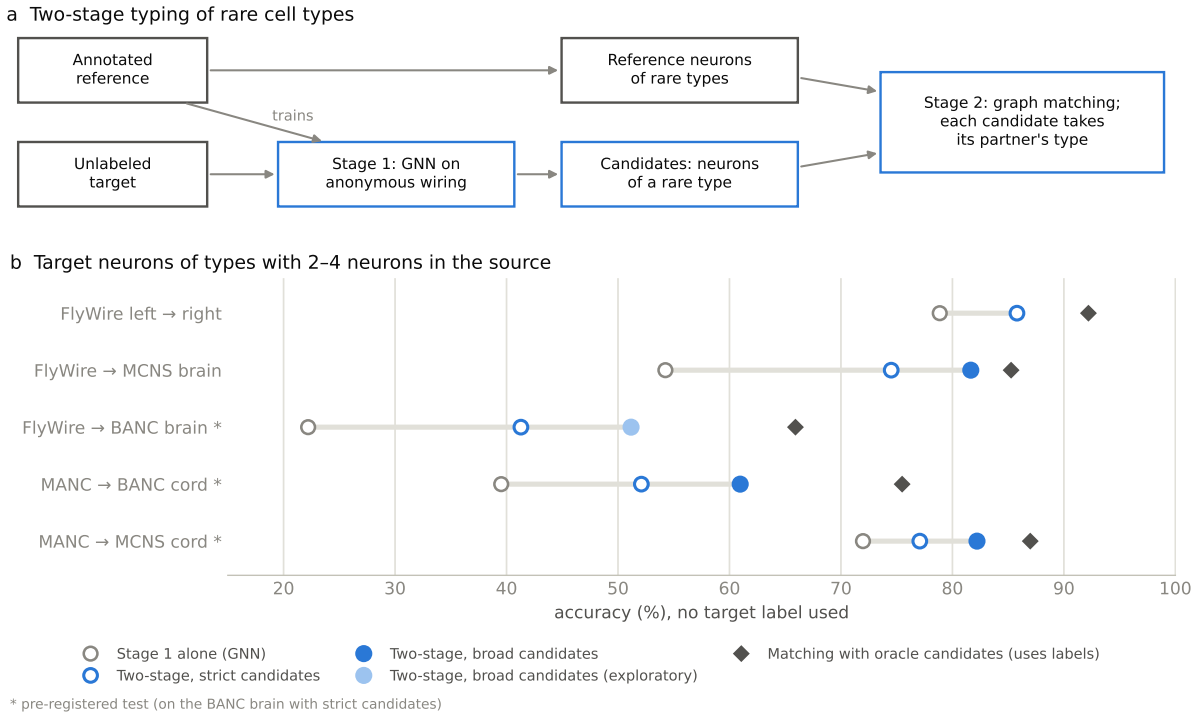


Figure 1: (a) The two-stage pipeline. A rare type has 2–4 neurons in the source. (b) Accuracy on the target neurons of rare types, without any target label, for each source → target pair. Stage 1 within FlyWire is the majority vote of three GNNs; elsewhere it is the label-free pipeline after NTAC (Section 3.2). Strict and broad candidates are defined in Table 2. Oracle candidates are the labeled rare-type target neurons, so that point uses labels and shows what perfect candidate selection would give. * pre-registered tests; on the BANC brain the pre-registered pipeline used strict candidates, and the broad-candidate point is exploratory. Numbers: Tables 3 and 4.

graph matching as the second stage after a transferred classifier has selected the candidates, across animals and connectomes.

Graph learning and domain adaptation. Neurons mostly connect to neurons of other types, a heterophilic setting in which standard message passing struggles (Zhu et al., 2020). Aggregating separately over in- and out-neighbors (Dir-GNN; Rossi et al., 2023) raises effective homophily; our model is Dir-SAGE (Hamilton et al., 2017; Rossi et al., 2023) with synapse-weighted means. For transfer without target labels, prior matching is a form of label-shift correction (Saerens et al., 2002) computed by Sinkhorn scaling (Cuturi, 2013), and it mirrors reference mapping in single-cell genomics, where batch effects separate a query from an annotated atlas. Our selection rule uses agreement between independently trained models, which tracks their error under distribution shift (Jiang et al., 2022).

3 Data and methods

3.1 Data, labels and references

Connectomes (Table 1). FlyWire FAFB v783 is the source for the brain and, with MCNS v1.0, one of the two development connectomes. BANC v888 and the hemibrain v1.2.1 were used only for testing: in pre-registered tests and, for BANC, afterwards in the analyses of Section 4.4. For the nerve-cord test, MANC v1.2.1 (harmonized by the BANC project) is the source and the nerve cords of BANC and MCNS are the targets; no nerve cord was used in development. Neurons are nodes; directed edges are connections weighted by synapse count. FAFB connections have at

Table 1: Connectomes. Conn./syn.: connections and synapses per neuron, counting connections of at least 5 synapses. Labeled neurons: brains, the 434 FlyWire types with at least 20 neurons; nerve cords, MANC types with at least 2 neurons. Test: used only for testing (Section 3.1).

Connectome	Role	Neurons	Conn./syn.	Labeled (types)	Notes
FAFB v783	source, development	139,255	26.8 / 364	108,647 (434)	female
MCNS v1.0 brain	development	144,725	35.9 / 494	109,901 (394)	male; different synapse detection
BANC v888 brain	tests	133,549	8.6 / 100	63,332 (409)	female; sparse; no R1-6 photoreceptors
Hemibrain v1.2.1	test	25,348	26.7 / 387	7,639 (202)	central brain only; 16 % of neurons cropped
MANC v1.2.1	source (cord)	23,650	55.1 / 969	22,883 (3,901)	male
BANC v888 cord	test	25,929	16.9 / 235	22,451 (3,677)	female; sparse
MCNS v1.0 cord	test	24,073	43.4 / 759	21,402 (3,670)	male

least 5 synapses (Codex); the other connectomes are aligned to the source by fixed, label-free rules (Supplement S1).

Rare types. The rare-type pipeline uses all FlyWire cell types with at least 2 neurons (8,658 types, 99 % of the labeled neurons). A **rare type** has 2–4 neurons in the source, and the **rare-type neurons** of a target are its labeled neurons of such types. In the nerve cord, labels are MANC cell types (3,901 types with at least 2 neurons, 2,452 of them rare). Stage 1 on common types (Section 4.5) uses the 434 FlyWire types with at least 20 neurons and 10 super classes.

References. In every target, labels are coded in the source's type names and used only for evaluation. The main reference, the **alignment reference**, is each target's own typing aligned to the source's types by its connectome team. How it was made matters for what agreement with it means. MCNS maps its neurons to FlyWire and MANC types by matching that uses connectivity. BANC matched its neurons to FlyWire and MANC neurons by NBLAST proposals that were reviewed by hand, partly with connectivity (Bates et al., 2026). On the rare-type neurons of BANC's brain, the reference type equals the type of the reviewed FlyWire match for 99.8 % of neurons, and that match is NBLAST's top hit for 67.5 %. BANC also has a **second reference**: the type of the morphology-matched FlyWire neuron (brain) and its curated typing (cord). For rare types the second references agree with the alignment references on 98–99.5 % of neurons, so they check the scoring but are not independent evidence. Every reference we have was built with morphology matching, connectivity or both.

3.2 Stage 1: a wiring-only GNN, transferred without labels

Each neuron has six features: in- and out-degree, input and output synapses, and mean synapses per input and output connection (log-scaled, z-scored). A layer computes $h_i' = W_{\text{self}}\, h_i + W_{\text{in}}\, \text{mean}_{\text{in}}(h) + W_{\text{out}}\, \text{mean}_{\text{out}}(h)$, with means over pre- and postsynaptic partners weighted by synapse count. The model never sees a partner's label, so it applies unchanged to an unlabeled graph. Three seeds are trained throughout (training details in Supplement S1).

A model trained on the source is applied to the target in three label-free steps:

- **Prior matching.** Cell-type sizes are stereotyped across flies, but a transferred classifier shifts neurons between neighboring types. We rescale its class probabilities so that the expected number of neurons per type matches the type distribution the same model predicts on its source, scaled to the target.
- **Choice of preprocessing.** How the target graph is aligned to the source, how its features

Table 2: Names used in this paper. `docs/reproduce.md` maps them to the code options.

Name	Meaning
Stage 1	The GNN trained on the source, applied to the target without labels: label-free choice of preprocessing, prior matching, NTAC refinement (Section 3.2).
Prior matching	Rescales the GNN's class probabilities so that predicted type sizes match the source's.
Size-and-agreement rule	Chooses stage 1's preprocessing without labels: best mean rank of type-size divergence from the prior and agreement between seeds.
Stage 2	Graph matching of the source's rare-type neurons to the candidates (Section 3.3).
Two-stage typing	Each matched candidate takes its partner's type; every other neuron keeps its stage-1 type.
Strict candidates	Neurons whose stage-1 type after NTAC is rare (the first rule).
Broad candidates	Neurons whose stage-1 type before or after NTAC is rare (the improved rule).
Morphology-augmented candidates	Broad candidates plus the neurons whose NBLAST match has a rare type.
Oracle candidates	The labeled rare-type target neurons; a diagnostic that uses labels.
Neck block	A third matching block for nerve-cord neurons that pass through the neck.
Seeded matching	Matching in which confident NBLAST matches are fixed in advance (Section 4.4).
Alignment / second reference	The target's typing aligned to the source's types / BANC's morphology-matched (brain) or curated (cord) typing (Section 3.1).

are normalized and how strongly prior matching is applied give 30 candidate preprocessings. The size-and-agreement rule chooses one without labels: the best mean rank of two properties, the divergence of the predicted type sizes from the prior and the agreement between the three seeds' labelings. It replaced an earlier rule after that rule failed on BANC (Section 4.5; Supplement S4).

- **Refinement.** NTAC on the target, seeded with the neurons on which all three seeds agree.

3.3 Stage 2: graph matching of rare types

- **Candidates** are the target neurons that stage 1 assigns to a rare type. The first tests used strict candidates (type after NTAC is rare). After the BANC brain test, the rule was improved on development data (Section 4.3): broad candidates take a neuron if its type before or after NTAC is rare.
- **Matching.** Stage 2 matches the source's rare-type neurons one-to-one to the candidates. It maximizes the agreement of the two log1p(synapse) adjacency matrices among the selected neurons by Frank–Wolfe on the relaxed quadratic assignment problem (FAQ; Vogelstein et al., 2015), from the barycenter, for 30 iterations. The assignment is block-diagonal by hemisphere and aligns both ipsilateral and contralateral connections (bisected matching; Pedigo et al., 2023). In the nerve cord a third, pooled neck block holds the ascending and descending neurons, whose side annotation follows the neck in MANC and the soma in the targets. Each block is padded to a square; candidates matched to padding are not assigned.
- **Final labeling.** A matched candidate takes the type of its source neuron; every other neuron keeps its stage-1 type.

At the barycenter the gradient is nearly of rank one and the exact assignment solver takes 5–10 minutes, so the first step pairs the leading singular vectors by rank. A block of about 9,300 neurons per side takes 27 minutes and 4.5 GB on a CPU, the nerve-cord blocks under 3 minutes. The GNN uses PyTorch Geometric (Fey and Lenssen, 2019), the assignment steps SciPy (Virtanen et al., 2020).

Table 3: Development data: accuracy % on the target neurons of rare types. Left → right: the 8,142 right-hemisphere FlyWire neurons, typed from the left hemisphere. FlyWire → MCNS: 13,742 MCNS neurons of 5,370 types. Stage 1 is the majority vote of three GNNs within FlyWire and the label-free pipeline after NTAC on MCNS. –: not run.

Method	FlyWire left → right	FlyWire → MCNS
Without any target label		
Stage 1	78.9	54.2
Stage-1 probabilities, made one-to-one on the strict candidates	–	59.9
Two-stage, strict candidates	**85.8**	74.5
Two-stage, broad candidates	–	**81.6**
Given which neurons are rare-type (oracle candidates)		
GNN votes, made one-to-one	79.8	–
Graph matching, ipsilateral connections only	91.2	–
Graph matching	92.2	85.3

3.4 Evaluation and pre-registration

Metrics. Accuracy is the share of labeled target neurons whose predicted type equals the reference type, computed over the rare-type neurons unless stated otherwise; a neuron without a predicted type counts as wrong. Macro-F1, the mean F1 over the true types present, is given in Supplement S6. The 95 % intervals resample types (2,000 draws; 10,000 for the analyses of Section 4.4), with the predictions fixed; they do not include the variability of the matching itself.

Pre-registration. For each test on a new dataset, the data, labels, candidates, selection rule, method, reported numbers and explicit expectations were committed to the repository before any model was run on it (Table 7). The protocols allow only data-loading and memory fixes afterwards and require every deviation to be reported. A fifth protocol fixed an analysis of wiring and morphology on BANC in the same way, but BANC's labels had been seen by then; we call it a pre-specified analysis, not a test on held-out data. Between tests, methods were changed only on degraded copies of the development connectomes, whose labels are known: synapse counts binomially thinned before the threshold, cell types removed, the volume restricted to the central brain, or a fraction of neurons cropped (each synapse of a cropped neuron kept with probability 0.1).

4 Results

4.1 Graph matching types rare neurons better than the GNN

Having seen one or two neurons of a type in the left hemisphere, the GNN types its right-hemisphere neurons at 75.9 % (mean of three seeds; majority vote 78.9 %). These types were defined by matching neurons across hemispheres (Schlegel et al., 2024), so the task is close to pairing homologs. We matched the left and right neurons of rare types (Section 3.3) and gave each right neuron the type of its matched left neuron (Table 3).

On the same candidates, graph matching beats the GNN by 12 points (92.2 vs. 79.8 %) and wins in every brain region, although it sees only the connections among about 8,000 neurons per side. The two methods combine without labels: the GNN's rare-type candidates contain 93 % of the true rare-type neurons (84 % precision), and matching them raises accuracy from 78.9 to 85.8 %.

The same division of labor works across animals. From FlyWire to MCNS, stage 1 types the rare-type neurons at 54.2 %. Its 18,479 strict candidates contain 84 % of them; matching these to the 16,365 FlyWire neurons of rare types gives 74.5 %, and broad candidates (92 % recall)

Table 4: Pre-registered tests: accuracy % on the target neurons of rare types, without any target label (alignment reference). The 95 % intervals of these accuracies are ±1.1–2.0 points. Gain: the pre-registered pipeline (bold) over stage 1, with 95 % interval. All intervals resample types with the predictions fixed; they do not include the training randomness of stage 1 (one run per target, combining three GNN seeds) or of the matching. † exploratory: run after the test.

Source; rare-type neurons	BANC brain FlyWire; 14,512	BANC cord MANC; 5,989	MCNS cord MANC; 5,945
Stage 1	22.2	39.5	72.0
Two-stage, strict candidates	**41.3**	52.1	77.1
Two-stage, broad candidates	51.2†	**61.0**	**82.2**
Oracle candidates	65.9	75.5	87.0
Gain over stage 1	+19.1 (18.0–20.1)	+21.4 (19.8–23.1)	+10.2 (8.9–11.5)
Pre-registered threshold	≥ 10: met	≥ 15: met	≥ 15: not met

give 81.6 %. The wiring does the work: making the GNN's probabilities one-to-one gains only 6 points, and starting the matching from them instead of the barycenter changes nothing (74.6 %). The largest gains are where the GNN is weakest: ascending, descending and other neurons rise from 25 to 66 %, central-brain neurons from 58 to 75 %. Matching between two animals is nearly as accurate as between two hemispheres (85.3 vs. 92.2 % with oracle candidates), so most of the remaining loss comes from candidate selection. Two caveats apply: stage 1 was developed on this pair, and MCNS's labels for these types come from cross-dataset matching that uses connectivity.

4.2 The gain holds on held-out connectomes

We tested the pipeline in three pre-registered tests (Table 4). On the BANC brain, the protocol (committed in `0d1ee67` before either stage ran on BANC at this label set) fixed strict candidates and the size-and-agreement rule, which chose the best of the 30 preprocessings. In the nerve cords (`5119069`), the source was MANC, on which the GNNs were trained unchanged; the targets were BANC's cord (female, 31 % of MANC's connections per neuron) and MCNS's cord (a second male). This test used the broad candidates and one change disclosed before any model output on a cord, the neck block.

The two stages beat stage 1 in all three targets. On the BANC brain, all four expectations held. Over all labeled neurons, macro-F1 rose from 0.19 to 0.40 and accuracy by 3.1 points, and graph matching withstood BANC's sparsity better than the GNN: with oracle candidates it kept 77 % of its MCNS accuracy, stage 1 only 41 %. In the nerve cords, two of four expectations held on both cords. Broad candidates beat strict ones by at least 3 points (+8.9 and +5.1), and the pipeline closed at least half of the gap between stage 1 and oracle candidates (0.60 and 0.68). The gain of 15 points held on BANC but not on MCNS, where stage 1 was already at 72 %. The fourth expectation, about a fallback for unmatched neurons, failed on both cords (Section 4.3).

Every neuron group gained on both cords; on BANC, descending neurons rose from 41 to 74 % and motor neurons from 16 to 48 %. Intrinsic neurons, whose references rest most on connectivity, gained no more than the others. Over all labeled neurons the pipeline also gained without the fallback (BANC 44.8 → 49.4 %, MCNS 73.1 → 75.0 %). BANC's second references give the same numbers (brain 21.3 → 40.6 %, cord 40.7 → 61.0 %), as expected from label sets that agree on 98–99.5 % of neurons.

4.3 Where the errors are: candidate selection

With oracle candidates, matching is 5–25 points more accurate than with the selected ones (Table 4). On the BANC brain, stage 1 finds only 54 % of the rare-type neurons (MCNS: 84 %),

Table 5: Candidate selection on development data: recall of the rare-type neurons and accuracy % on them, MCNS thinned to 25 % of its synapses and full.

Candidates	Recall, thinned	Accuracy, thinned	Accuracy, full MCNS
Strict (as tested on the BANC brain)	54.4	47.3	74.5
Broad	70.0	**60.1**	**81.6**
Oracle	100	82.0	85.3

while matching types 77 % of those it finds correctly. Because BANC was used up for rare types after that test, we improved candidate selection on development data only. MCNS thinned to 25 % of its synapses reproduces BANC's stage 1 (candidate recall 54.4 vs. 54.3 %, accuracy 19.2 vs. 22.2 %) and served as a stand-in (Table 5).

Selecting more candidates helps. Broad candidates raise recall from 54 to 70 % on the thinned copy and from 84 to 92 % on full MCNS, and matching stays accurate on the added neurons (81 %). The rule has no free parameter, and a second thinning realization agrees within 1–2 points. On the BANC brain, broad candidates later raised accuracy from 41.3 to 51.2 % (exploratory), and adding the neurons whose NBLAST match has a rare type raised candidate recall to 95 % and accuracy to 62.6 % (Section 4.4).

Three other changes did not help. Letting a source neuron match two candidates failed on development data (oracle candidates: 85.3 → 76.1 %), because candidates piled onto a few well-connected neurons. A coarse fallback, which typed unmatched neurons with a model trained on types of at least 10 neurons, added 2.5–3.7 points over all labeled neurons on development data, but cost 0.4 and 1.2 points on the cords; we dropped it. The neck block did help: without it, the cords' rare types lost 5.6–5.7 points. The cost of the matching is borne by types of 5–19 neurons (−1.5 to −2.5 points), because a fifth to a third of the candidates belong to other types and receive a rare type.

4.4 Wiring and morphology

New reconstructions come with neuron shapes, and current typing starts from morphology matches (NBLAST; Costa et al., 2016). BANC's metadata holds each neuron's NBLAST top match to FlyWire and to MANC before review, so we could compare and combine the two signals on BANC (Table 6, Figure 2).

Exploratory comparison. On the brain, morphology alone beats wiring alone (66.5 vs. 51.2 %), but the two err on different neurons. Where NBLAST and the pre-registered two-stage typing agree, both are right in 95 % of cases; where they disagree, NBLAST is right in 61 % and the matching in 16 %. We tried six label-free combinations (planned in `docs/finalize_plan.md`, commit `9ecc769`, but run on a target whose labels were known). The best one, which takes the matched type where the matching assigns a FlyWire neuron and NBLAST's type elsewhere, reached 72.5 %, 6.0 points above NBLAST alone. In the cord, NBLAST's top hit has the reviewed match's type in only 31 % of cases, and wiring alone is twice as accurate.

Pre-specified analysis. Because the +6.0 was the best of six exploratory combinations, we fixed one further procedure, its contrast and its decision rule before running it (`docs/preregistration_confirmatory.md`, commit `3949b10`). We fixed a new procedure rather than one of the six, including the second best (70.6 %), because their results were already known: registering one of them would not have removed the selection concern. We chose seeding to test whether confident morphology matches can anchor the matching. A confident NBLAST match, meaning a candidate whose NBLAST top hit is a rare-type FlyWire neuron in the same hemisphere and the top hit of no other candidate, is fixed as a seed (seeded graph matching; Fishkind et al., 2019; 6,138 seeds in the brain). The matching then pairs the remaining candidates

Table 6: Wiring and morphology on BANC: accuracy % on the target neurons of rare types, without any BANC label (alignment reference). NBLAST: the type of the top morphology match, before review. Combination: the matched type where the matching assigns a source neuron, NBLAST's type elsewhere. The exploratory rows were chosen after BANC's labels were known; the pre-specified analysis was fixed before it was run, but on the same labels. The interval resamples types with the predictions fixed.

Labeling	BANC brain (14,512)	BANC cord (5,989)
NBLAST alone (morphology)	66.5	30.5
Two-stage, broad candidates (wiring only)	51.2	61.0
Two-stage, morphology-augmented candidates	62.6	64.7
Exploratory combinations		
Combination, broad candidates (best of six)	72.5	64.9
Combination, morphology-augmented candidates	70.6	65.1
Combination, NBLAST-only candidates (no GNN; run last)	58.9	45.6
Pre-specified analysis: seeded matching (labels seen before)		
Two-stage, seeded	61.6	55.4
Combination, seeded	**68.7**	**53.7**
Gain over unreviewed NBLAST (95 % interval)	+2.3 (1.3–3.2)	+23.2 (21.8–24.5)
Rule: gain $\geq$ 3 points, interval above 0	not met	met (secondary)

around the seeds, and each neuron takes the matched type where the matching assigned one and NBLAST's type elsewhere. The rule called the gain confirmed if it reached at least 3 points over NBLAST alone with a 95 % interval above zero. On the brain, the combination reached 68.7 % against 66.5 % for NBLAST: a gain of 2.3 points (1.3–3.2), below the bar. Against the second reference the gain was 1.4 points (0.4–2.4). Wiring therefore adds to morphology on the brain, but by less than the margin we required, and the exploratory +6.0 is not confirmed. In the cord, the combination beat NBLAST by 23.2 points. BANC's labels had been seen before this analysis, so it is fixed in procedure, not tested on unseen data.

Seeding itself hurt. Without seeds, on the same candidates, the combination reached 70.6 % (brain) and 65.1 % (cord) instead of 68.7 and 53.7 %. About 16 % of the brain seeds and 32 % of the cord seeds carry a wrong type, and fixing them prevents the matching from correcting them.

Is the GNN still needed when morphology is available? To find out, we ran the matching without stage 1 (exploratory, after the pre-specified analysis). The candidates were only the neurons whose NBLAST match has a rare type; they contain 85 % of the brain's rare-type neurons and 63 % of the cord's, against 95 and 90 % when the GNN's candidates are added. Each neuron took the matched type where the matching assigned one and NBLAST's type elsewhere, so no step used the GNN. Neurons that neither typed, exactly those without an NBLAST hit (7.1 % of the brain's rare-type neurons, 1.2 % of the cord's), count as wrong, as they do for NBLAST alone. This GNN-free combination reached 58.9 % on the brain, 7.6 points below NBLAST alone, and 45.6 % in the cord. With the GNN's candidates added, the same combination reached 70.6 and 65.1 %: the GNN contributes 11.7 points on the brain (95 % interval 11.0–12.4) and 19.5 in the cord (18.2–20.8). Of these, 4.3 and 0.8 points come from the neurons without an NBLAST hit, which the version with the GNN types by stage 1; the rest comes from neurons that NBLAST does match. A likely reason is that one-to-one matching on a candidate set that lacks many true partners forces wrong pairs, and the GNN supplies the missing neurons.

The reference favors NBLAST. BANC's types were assigned by reviewing and correcting NBLAST proposals (Section 3.1). NBLAST's accuracy against them is therefore largely the share of top hits that reviewers kept: it is right on 99.9 % of the neurons where review kept its top hit and on 12.0 % where review replaced it (Figure 2b). Where reviewers replaced the hit, the pre-specified combination recovers the reviewed type for 39.1 % of the neurons and the

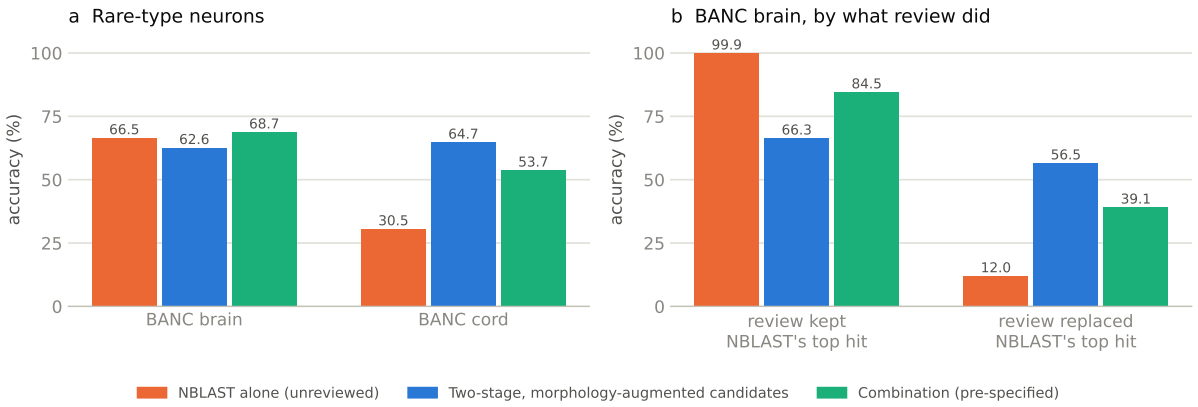


Figure 2: Wiring and morphology on BANC's rare-type neurons (alignment reference). (a) NBLAST alone, the two-stage typing with morphology-augmented candidates, and the pre-specified combination (seeded matching, NBLAST's type where the matching assigns none). NBLAST: unreviewed top matches. (b) The brain neurons split by whether BANC's reviewers kept NBLAST's top hit as the match (9,002 neurons) or replaced it (4,331). Where review kept it, the reference is NBLAST's own answer.

unseeded two-stage typing for 56.5 %. Where they kept it, the combination loses 15.4 points against a reference that is NBLAST's answer by construction. Review also used connectivity, which may favor wiring where reviewers overrode NBLAST. Comparisons between wiring and morphology on BANC are therefore tilted, mostly toward morphology. A cleaner test would need a target whose types were assigned independently of NBLAST; MCNS, the other target, has no automatic morphology match to FlyWire. Over all labeled brain neurons, the pre-specified combination reached 61.0 % against 56.4 % for NBLAST.

4.5 Stage 1 on its own: common types

The pipeline relies on stage 1 transferring without labels. We developed and tested it first on the 434 FlyWire types with at least 20 neurons; the Supplement gives the details. On an unlabeled hemisphere, where the median neuron has no synapse to a labeled partner, methods that read partner labels collapse on cell types (1–9 %), while the GNN keeps 87–95 %. Applied to MCNS, stage 1 types 394 cell types at 92.6 % without a single MCNS label, against 96.6 % for a GNN trained on MCNS; most of this is the optic lobe (97.2 %; other neurons 74.0 %). On the sparse BANC brain it reached 61.7 %, and density explains about half of the drop. The first rule for choosing the preprocessing failed there (for MCNS → BANC it chose a labeling that put 77 % of the neurons into one type) and was replaced by the size-and-agreement rule, which chose the best or a near-best preprocessing in every later test except the partial volume. A partial volume, the hemibrain, cannot be typed from wiring: cell types reached 19.6 % and super classes 74.4 %, barely above predicting the most frequent class (73.1 %). On complete targets, a few percent of hand labels propagated by NTAC type common types better; the GNN is worth about 1.3 % hand labels.

5 Discussion

Pairing beats classifying for rare types. A classifier has seen one or two examples of a rare type, while graph matching uses the wiring among all candidates at once. Within FlyWire, on the same candidates, matching wins by 12 points; between animals, the transferred GNN's role is to decide which neurons to match. This division of labor held in three pre-registered tests: across sex, at a third of the source's density, and from brain to nerve cord with a new source. These

are also the types a hand-labeling budget reaches last. Candidate selection is the bottleneck, and the GNN remains needed for it when morphology matches exist: without its candidates, matching combined with NBLAST fell below NBLAST alone on the brain. One-to-one matching ignores that type sizes differ between animals; a looser assignment failed, because candidates piled onto a few well-connected neurons.

Wiring next to morphology. In practice a new reconstruction is typed from morphology first, and our tests support a narrower claim than the exploratory analysis suggested. Wiring complements morphology: the two err on different neurons. We did not show that they are statistically independent, and our references are built from both. On BANC's brain, where unreviewed NBLAST matches are good, wiring adds little to them: 2.3 points in the pre-specified analysis, which missed its 3-point rule, and up to 6 points in exploratory combinations. In BANC's cord, where these matches confuse sibling types and classes, it adds a lot (23 points over unreviewed NBLAST, not over reviewed morphology matching). Where wiring and morphology agree, the type is almost always right (95 %), which gives a cheap confidence flag for review, and morphology in turn fixes the wiring pipeline's bottleneck by raising candidate recall from 70 to 95 %. Seeding the matching with hard NBLAST matches hurt; using NBLAST as a soft prior term in the matching is the natural next step. The size of the effect on BANC is uncertain in both directions, because the reference is partly NBLAST's own answer.

Circularity and limitations. Cell types were defined partly from connectivity, and every cross-dataset reference we used comes from matching that uses connectivity, morphology or both. Agreement with them is therefore partly agreement with another matching procedure. The GNN never sees partner identities, which weakens but does not remove this concern, and intrinsic nerve-cord neurons, whose references rest most on connectivity, gained no more than others. BANC's second references are not independent of its alignment references, and its NBLAST-based labels favor NBLAST. The rare-type pipeline was developed on one brain pair and tested on one held-out brain and two nerve cords. Each target was run once (three GNN seeds combined into one labeling), so variation between such runs is not in our intervals. Single-neuron types are not predicted, and serial homologs in the nerve cord are not separated by the matching blocks. The recommended pipeline drops the fallback after the last test, so its all-neuron numbers come from runs reported in that test, not from a fresh one. The setting is transductive within a connectome. The size-and-agreement rule was chosen on 14 degraded development cases. Thinned copies cannot contain connections below the 5-synapse threshold. Results depend on how neurons and classes are aligned across datasets.

On pre-registration (Table 7). The first two tests contradicted part of what development data suggested, and the third confirmed it. The fourth confirmed the core method but rejected an add-on tuned on development data, and it showed that an absolute threshold can be out of reach when the baseline is already strong; thresholds relative to the oracle are the better form. The fifth, a pre-specified analysis on labels we had seen, turned an exploratory +6.0 into a smaller gain that missed its bar. Degraded copies of annotated connectomes let us diagnose and repair failures without spending a new dataset; we recommend them as a test bed for any method meant to type new reconstructions.

6 Conclusion

Rare cell types, most of a fly connectome's types, can be typed without any target label by letting a transferred wiring-only GNN select the candidates and graph matching type them. In pre-registered tests the approach roughly doubles accuracy on a held-out brain (22 → 41 %) and, with broader candidate selection, carries over to two nerve cords from a new source (40 → 61 % and 72 → 82 %). Wiring complements morphology: combined with unreviewed automatic morphology matches, it adds little where these matches are good (+2.3 points on BANC's brain,

Table 7: The five protocols fixed in advance, in the order they were run. The first four are tests on data not used before; the fifth is a pre-specified analysis on BANC, whose labels had been seen.

Test	Protocol (commit)	What was tested	Expectations
BANC brain, common types	`banc_protocol.md` (`85516a2`)	unchanged stage 1, first selection rule	3: one held in every run, two only in some runs
Hemibrain	`hemibrain_protocol.md` (`161cf7b`)	size-and-agreement rule, partial volume	4: three held, one for super classes only
BANC brain, rare types	`banc_rare_protocol.md` (`0d1ee67`)	two-stage pipeline, strict candidates	4: all held
Nerve cords, rare types (MANC → BANC, MCNS)	`vnc_rare_protocol.md` (`5119069`)	broad candidates, fallback, neck block	4 per cord: two held on both, one on BANC only, the fallback failed on both
BANC, wiring and morphology (pre-specified; labels seen before)	`preregistration_confirmatory.md` (`3949b10`)	seeded matching combined with NBLAST	1: not met on the brain ($+2.3 < 3$ points)

short of the bar we fixed in advance) and a lot where they are weak (+23 points in BANC's cord). Candidate selection is the bottleneck, sparser reconstructions cost accuracy, and partial volumes are out of reach for wiring alone. The approach is most useful for the rare types that a labeling budget reaches last.

Code and data availability

Code, configs, protocols and the commands for every number are in the repository (https://github.com/Paradoxsmile/flybrain; archived on Zenodo: doi:10.5281/zenodo.23768609); `docs/reproduce.md` maps each table to its command and result file, and the small result summaries are committed under `results/`. Data are public (FlyWire Codex, Janelia MCNS, BANC, and the hemibrain and MANC via the BANC project's bucket). Protocols: `docs/banc_protocol.md` (commit `85516a2`), `docs/hemibrain_protocol.md` (`161cf7b`), `docs/banc_rare_protocol.md` (`0d1ee67`), `docs/vnc_rare_protocol.md` (`5119069`), `docs/preregistration_confirmatory.md` (`3949b10`).

References

Alexander S. Bates, Jasper S. Phelps, Minsu Kim, Helen H. Yang, et al. Distributed control circuits across a brain-and-cord connectome. *Nature*, 2026. doi: 10.1038/s41586-026-10735-w. bioRxiv 10.1101/2025.07.31.667571.

Sophie Berg et al. Sexual dimorphism in the complete connectome of the *Drosophila* male central nervous system. *Cell*, 189(18):5504–5526, 2026. doi: 10.1016/j.cell.2026.08.015. bioRxiv 10.1101/2025.10.09.680999.

Marta Costa, James D. Manton, Aaron D. Ostrovsky, Steffen Prohaska, and Gregory S. X. E. Jefferis. NBLAST: rapid, sensitive comparison of neuronal structure and construction of neuron family databases. *Neuron*, 91(2):293–311, 2016.

Marco Cuturi. Sinkhorn distances: lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2013.

Sven Dorkenwald et al. Neuronal wiring diagram of an adult brain. *Nature*, 634:124–138, 2024.

Matthias Fey and Jan Eric Lenssen. Fast graph representation learning with PyTorch Geometric. In *ICLR Workshop on Representation Learning on Graphs and Manifolds*, 2019.

Donniell E. Fishkind, Sancar Adali, Heather G. Patsolic, Lingyao Meng, Digvijay Singh, Vince Lyzinski, and Carey E. Priebe. Seeded graph matching. *Pattern Recognition*, 87:203–215, 2019.

William L. Hamilton, Rex Ying, and Jure Leskovec. Inductive representation learning on large graphs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Yiding Jiang, Vaishnavh Nagarajan, Christina Baek, and J. Zico Kolter. Assessing generalization of SGD via disagreement. In *International Conference on Learning Representations (ICLR)*, 2022.

Minghui Liao, Guojia Wan, and Bo Du. Joint learning neuronal skeleton and brain circuit topology with permutation invariant encoders for neuron classification. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 197–205, 2024.

Elizabeth C. Marin et al. Systematic annotation of a complete adult male *Drosophila* nerve cord connectome reveals principles of functional organisation. *eLife*, 13:RP97766, 2024.

Arie Matsliah et al. Neuronal parts list and wiring diagram for a visual system. *Nature*, 634, 2024.

Benjamin D. Pedigo, Michael Winding, Carey E. Priebe, and Joshua T. Vogelstein. Bisected graph matching improves automated pairing of bilaterally homologous neurons from connectomes. *Network Neuroscience*, 7(2):522–538, 2023. doi: 10.1162/netn_a_00287.

Carey E. Priebe, Youngser Park, Minh Tang, Avanti Athreya, Vince Lyzinski, Joshua T. Vogelstein, Yichen Qin, Ben Cocanougher, Katharina Eichler, Marta Zlatic, and Albert Cardona. Semiparametric spectral modeling of the *Drosophila* connectome, 2017. arXiv:1705.03297.

Emanuele Rossi, Bertrand Charpentier, Francesco Di Giovanni, Fabrizio Frasca, Stephan Günnemann, and Michael M. Bronstein. Edge directionality improves learning on heterophilic graphs. In *Learning on Graphs Conference (LoG)*, 2023.

Marco Saerens, Patrice Latinne, and Christine Decaestecker. Adjusting the outputs of a classifier to new a priori probabilities: a simple procedure. *Neural Computation*, 14(1):21–41, 2002.

Louis K. Scheffer et al. A connectome and analysis of the adult *Drosophila* central brain. *eLife*, 9:e57443, 2020.

Philipp Schlegel et al. Whole-brain annotation and multi-connectome cell typing of *Drosophila*. *Nature*, 634:139–152, 2024.

Gregory Schwartzman, Ben Jourdan, David García-Soriano, and Arie Matsliah. NTAC: Neuronal type assignment from connectivity. *Nature Communications*, 17, 2026. doi: 10.1038/s41467-025-68044-1.

Shin-ya Takemura et al. A connectome of the male *Drosophila* ventral nerve cord. *eLife*, 13: RP97769, 2024.

Jakob Troidl et al. Global neuron shape reasoning with point affinity transformers, 2024. bioRxiv.

Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020.

Joshua T. Vogelstein, John M. Conroy, Vince Lyzinski, Louis J. Podrazik, Steven G. Kratzer, Eric T. Harley, Donniell E. Fishkind, R. Jacob Vogelstein, and Carey E. Priebe. Fast approximate quadratic programming for graph matching. *PLoS ONE*, 10(4):e0121002, 2015.

Jiong Zhu, Yujun Yan, Lingxiao Zhao, Mark Heimann, Leman Akoglu, and Danai Koutra. Beyond homophily in graph neural networks: current limitations and effective designs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

Supplementary material

Sections S1–S5 give the details of stage 1 and the common-type results that Section 4.5 summarizes; they use the 434 FlyWire types with at least 20 neurons unless stated otherwise. Section S6 gives macro-F1 for the rare-type tables of the main text.

S1 Details of stage 1

Training. Cell types use 3 layers (300 epochs, class-weighted loss), super classes 4 layers (1,000 epochs); depth and options were chosen by validation macro-F1. With thousands of types the output layer and prior matching are computed in chunks, which reproduces the standard code where both can run.

Alignment variants. Connectomes differ in synapse detection and completeness. Each target gets variants that align one statistic with the source by a fixed rule: *unaligned* (threshold 5, nothing aligned), *density-aligned* (the threshold whose connections per neuron are closest to the source's) and *count-scaled* (threshold 5, counts scaled so that synapses per neuron match the source's). For MCNS against FAFB we use Janelia's published scaling factor (0.581) for the last rule. In the nerve-cord test, self-connections and connections between two neurons that pass through the neck, which in the targets can lie in the brain, are dropped in all three graphs.

Candidate preprocessings. Each of the three alignment variants is combined with two feature normalizations (the source's training statistics or the target's own) and five strengths of prior matching, $\rho \in \{0, 0.25, 0.5, 0.75, 1\}$: 30 candidates.

Prior matching. We rescale the class probabilities with weights v (Sinkhorn iterations) so that the expected number of neurons per type matches the type distribution the same model predicts on its source, scaled to the target; ρ interpolates between the raw prediction and full matching ($p_k \cdot v_k^\rho$, renormalized). For partial volumes, a coverage prior uses the type distribution predicted for the FAFB neurons inside the covered volume, those with at least 10 % of their synapses outside the optic-lobe neuropils; it needs only the list of covered neuropils.

Selection rules. The *type-consistency rule*, used in the first test, picks the preprocessing whose labeling (mean over three seeds) is most self-consistent on the target: each neuron of a random half gets the label of its most similar neuron in the other half (synapses per partner type, weighted Jaccard), and agreement is chance-corrected (Cohen's κ). The *size-and-agreement rule*, which replaced it after the first test and is used in all rare-type tests, takes the best mean rank of the Jensen-Shannon divergence of the predicted type sizes from the prior and the agreement between the three seeds' labelings.

S2 Stage 1 within a connectome

Baselines, ordered by the information they use: majority class; logistic regression on a neuron's own features; label propagation; a partner profile (logistic regression on the share of synapses per partner type, using training labels); NTAC (Schwartzman et al., 2026; a memory-lean reimplementation checked step by step against the published package); GraphSAGE (inputs only, unweighted). Splits: random (70/15/15, stratified) and hemisphere (train and validate on one hemisphere, test on the other). Three seeds throughout (Figure S1, Table S1).

With labeled partners, NTAC is the best cell-type method; the GNN comes close (95.3 vs. 97.0 %, equal macro-F1) without seeing any partner label and is best on super classes. On an unlabeled hemisphere, the median test neuron has no synapse to a labeled partner (optic-lobe circuits stay on their side), and partner-label methods collapse on cell types (1–9 %; NTAC stays below 9.4 %

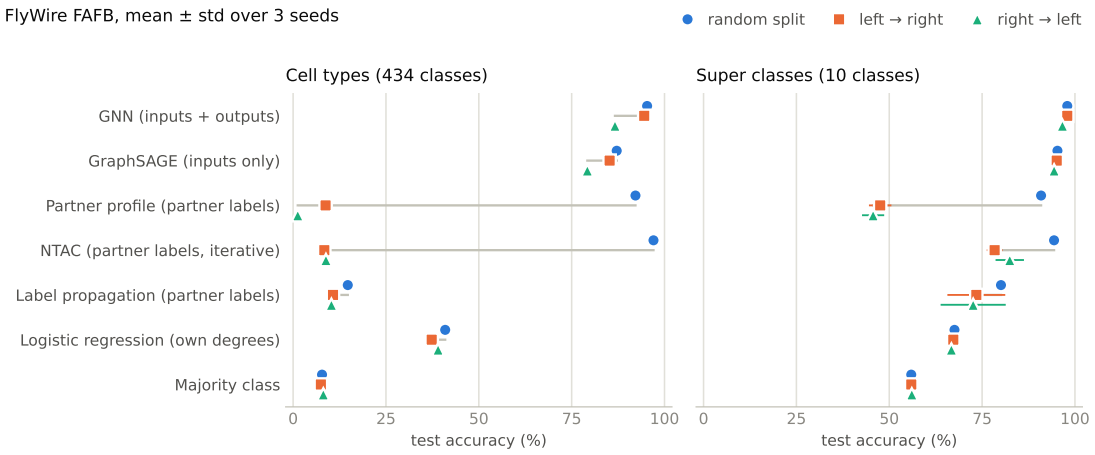


Figure S1: Test accuracy on FlyWire per method and split (mean over 3 seeds, whiskers: std).

Table S1: FlyWire, test accuracy % / macro-F1 (mean ± std, 3 seeds).

Method	Cell types, random	left → right	right → left	Super classes, random	left → right	right → left
Majority class	7.8 / 0.00	7.5 / 0.00	8.1 / 0.00	55.9 / 0.07	55.9 / 0.07	56.1 / 0.07
Logistic regression (own degrees)	40.9 / 0.12	37.3 / 0.09	39.0 / 0.11	67.6 / 0.31	67.2 / 0.32	66.7 / 0.32
Label propagation	14.7 / 0.28	10.7 / 0.19	10.3 / 0.20	80.1 / 0.30	73.4 / 0.22	72.6 / 0.21
Partner profile (partner labels)	92.2 / 0.79	8.7 / 0.10	1.2 / 0.10	90.9 / 0.60	47.6 / 0.28	45.7 / 0.23
NTAC (partner labels, iterative)	**97.0 / 0.93**	8.4 / 0.20	8.8 / 0.19	94.4 / 0.78	78.4 / 0.49	82.4 / 0.58
GraphSAGE	87.1 / 0.77	85.2 / 0.72	79.2 / 0.69	95.3 / 0.62	95.1 / 0.63	94.4 / 0.63
GNN (Dir-SAGE, weighted)	95.3 / **0.93**	**94.5 / 0.90**	**86.7 / 0.87**	**97.9 / 0.85**	**97.9 / 0.84**	**96.6 / 0.76**

at any of 40 steps), while the GNN keeps 87–95 %. Aggregating over outputs as well as inputs adds 2–9 points over GraphSAGE of the same depth. Cell types are strongly heterophilic: only 3 % of synapses connect neurons of the same type. The right → left gap of 8 points concentrates on photoreceptors and lamina neurons, whose outgoing synapses are underdetected in FlyWire (Matsliah et al., 2024) and whose connection counts differ between the hemispheres.

The result replicates within MCNS: the GNN reaches 96.6 % (random), 95.4 % and 93.7 % (hemispheres) on cell types, NTAC 7.8–9.5 % on the hemispheres. Seeding NTAC with the predictions on which three GNNs agree yields 96.5 / 90.8 % (FlyWire) and 97.1 / 96.2 % (MCNS) on the unlabeled hemisphere.

S3 Stage 1 between complete connectomes, down to types of two neurons

A single FlyWire GNN applied to MCNS with a label-free normalization reaches 73–75 % on cell types; the best alignment, found only post hoc, 85.7 %. The label-free pipeline chooses the alignment and normalization with the type-consistency rule, which ranks the 30 candidates like their accuracy (Spearman 0.95–0.97), and prior matching corrects the type sizes: 87.6 % without and 91.2 % with prior matching, and **92.6 % after NTAC, over 394 cell types without a single MCNS label**, against 96.6 % for a GNN trained on MCNS (Figure S2). Prior matching

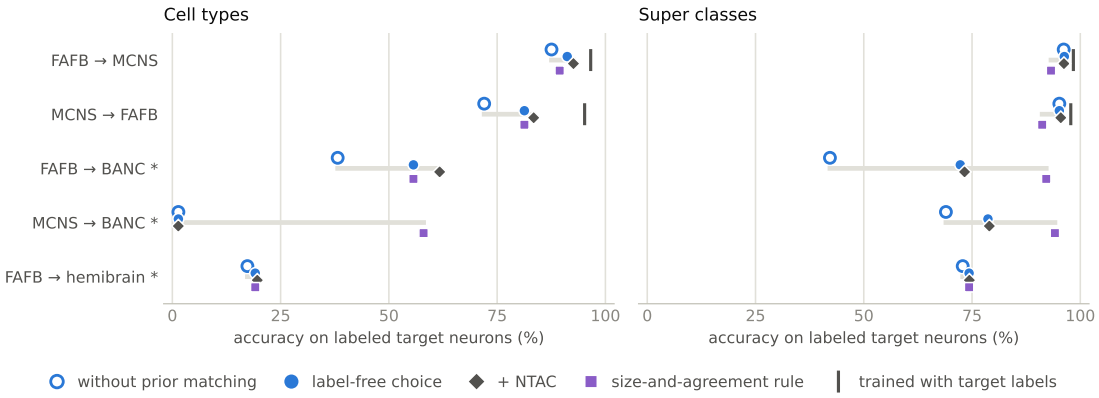


Figure S2: Label-free transfer per scenario (accuracy on labeled target neurons). Open circles: chosen preprocessing without prior matching; filled circles: the label-free choice with prior matching; diamonds: after NTAC; squares: the size-and-agreement rule; bars: a GNN trained with labels on the target. * pre-registered tests; their filled circles and diamonds use the rule fixed in the protocol (type consistency for BANC, size and agreement for the hemibrain).

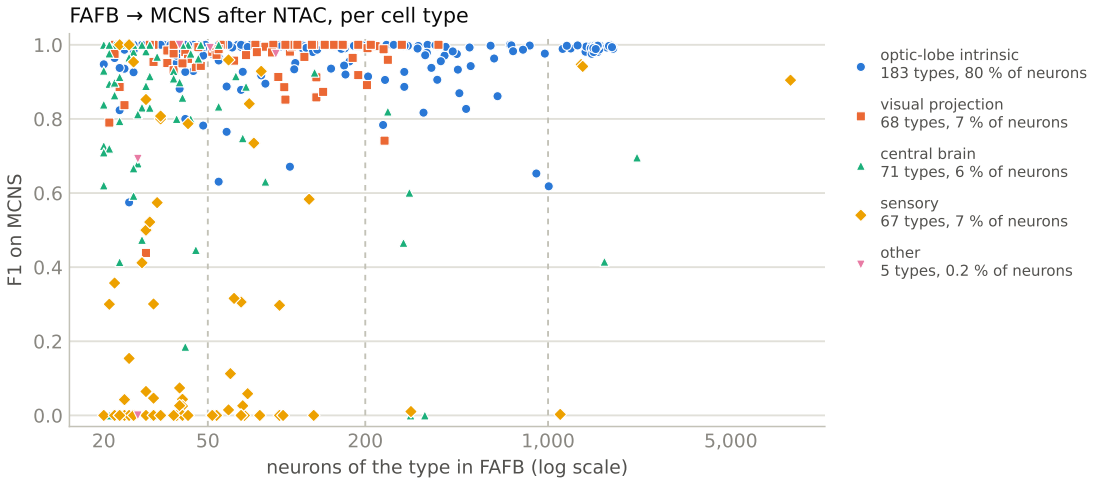


Figure S3: FAFB → MCNS after the label-free pipeline (+ NTAC): F1 per cell type on MCNS against the type's number of neurons in FlyWire, by region (FlyWire super class of the type). Dashed lines: the type-size bins of the stratified table.

repairs the types the transfer over- or under-fills (T5a: 2,252 → 1,714 predicted neurons for 1,664 true ones; LC21: 13 → 115 for 153). Run unchanged in the reverse direction, the pipeline reaches 83.4 % on FlyWire (72.0 % without prior matching; 95.2 % trained within FlyWire). The larger gap is specific to FlyWire as a target: 2.2 % of its neurons belong to types absent from MCNS's mapping, photoreceptors cause 21 % of the errors (51 % correct in the left hemisphere, 83 % in the right), and Kenyon cell subtypes 15 %.

Where the accuracy comes from. The headline number is mostly the optic lobe (Figure S3). 80 % of MCNS's labeled neurons belong to optic-lobe intrinsic types; they are typed at 97.2 % after NTAC and contribute 78.0 of the 92.6 points, while the other 20 % are typed at 74.0 %. Central-brain types transfer much worse (65.1 %), mostly because of Kenyon cell subtypes (84 % of their errors). The pattern repeats in the reverse direction (optic lobe 92.6 %, all other neurons 60.9 %) and on BANC (68.6 % and 44.9 %). Prior matching helps central-brain types most (+11, +16 and +26 points in the three runs).

Smaller types. We retrained the GNN with minimums of 10, 5, 3 and 2 neurons per type, everything else unchanged, and repeated the label-free FAFB → MCNS transfer (Table S2). The

Table S2: Cell types by minimum type size, accuracy % / macro-F1 (macro-F1: mean over the true types present). FlyWire: mean ± std over 3 seeds. Transfer: label-free pipeline + NTAC.

Min. size	Types (FlyWire neurons)	FlyWire, left → right	FAFB → MCNS (MCNS types)	MCNS central-brain types	Rarest bin: left → right / transfer
20	434 (78 %)	94.3 ± 0.7 / 0.92	92.6 / 0.81 (394)	65.1 / 0.80 (71)	20–49: 90.7 / 77.0
10	779 (81 %)	93.1 ± 1.1 / 0.87	**93.0 / 0.80** (649)	77.7 / 0.79 (243)	10–19: 86.6 / 75.5
5	2,164 (87 %)	89.1 ± 0.9 / 0.77	90.3 / 0.71 (1,694)	72.3 / 0.70 (1,181)	5–9: 76.4 / 71.6
3	4,175 (93 %)	87.2 ± 0.8 / 0.72	87.6 / 0.57 (3,206)	62.5 / 0.54 (2,542)	3–4: see ≥ 2
2	8,658 (99 %)	83.8 ± 1.1 / 0.71	84.2 / 0.51 (7,064)	64.3 / 0.54 (5,371)	2–4: 75.9 / 54.2

Table S3: BANC, accuracy % / macro-F1, as pre-registered (type-consistency rule).

Run, label	Without prior matching	With prior matching	+ NTAC	Best candidate (post hoc)
FAFB → BANC, cell types	38.2 / 0.21	55.7 / 0.37	**61.7 / 0.40**	55.7
FAFB → BANC, super classes	42.2 / 0.11	72.3 / 0.24	**73.3 / 0.22**	92.6
MCNS → BANC, cell types	1.4 / 0.00	1.4 / 0.00	**1.4 / 0.00**	58.0
MCNS → BANC, super classes	69.0 / 0.34	78.7 / 0.33	**79.0 / 0.31**	94.3

≥ 2 run is stage 1 of the rare-type pipeline.

Lowering the minimum from 20 to 10 costs nothing after transfer: no type-size bin is worse (20–49 neurons: 80.4 vs. 77.0 %), and 255 types are added. Below 10, each step trades accuracy on common types for coverage: at ≥ 2 the 20–49 bin falls to 73.7 % and the largest types (≥ 1,000 neurons) from 95.1 to 92.6 %. With 243 central-brain types at ≥ 10, central-brain accuracy rises to 77.7 %, and with thousands of rare ones it settles at 62–64 %. Optic-lobe and visual projection types stay at 93–97 % at every threshold, while small ascending, descending and other types fall to 32 % at ≥ 2. Prior matching matters more with many types (+13.2 points at ≥ 2, +3.6 at ≥ 20). NTAC still adds 3–4 points of accuracy but lowers macro-F1 at ≥ 3 and ≥ 2 (0.62 → 0.57, 0.59 → 0.51), because it pulls neurons of rare types into common ones. Rare types transfer, but with a widening gap: types of 10–19 FlyWire neurons are typed in MCNS at 75.5 %, of 5–9 neurons at 71.6 % (1,045 types) and of 2–4 neurons at 54.2 % (5,370 types, 13,742 neurons), against 86.6, 76.4 and 75.9 % within FlyWire across hemispheres.

S4 Stage 1 on sparse and partial targets

BANC brain, common types. The first pre-registered test applied the unchanged ≥ 20 pipeline to BANC (`docs/banc_protocol.md`, `85516a2`), with the type-consistency rule (Table S3).

Of three pre-registered expectations, the first (selected candidate within 2 points of the best) held in one of four runs; the second (the pipeline beats every candidate without prior matching on cell types) held for FAFB → BANC only; the third (cell-type accuracy below FAFB → MCNS) held. Prior matching carried over: the best cell-type candidates use full prior matching, 13.0 and 17.5 points above the best candidates without it, although BANC lacks the R1-6 photoreceptors that

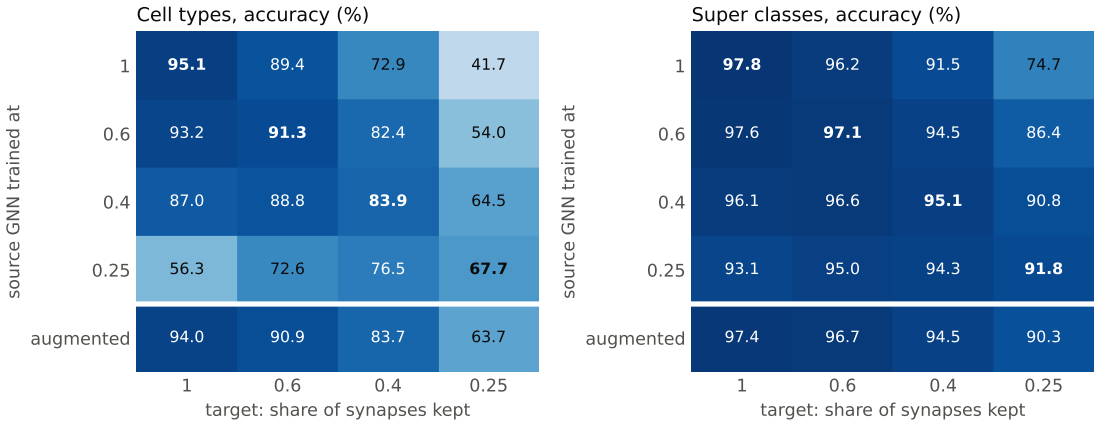


Figure S4: Transfer between thinned FlyWire copies, with prior matching (test split, mean over 3 seeds). Rows: density of the source copy the GNN was trained on; "augmented": one GNN trained with a random density in [0.25, 1] per epoch.

the prior expects (prior matching fills R1-6 with 10,800 neurons, 20 % of the remaining errors). The selection did not: for MCNS → BANC cell types it chose a labeling that puts 77 % of the neurons into one type; such a labeling is self-consistent on the graph, and the chance correction does not penalize the collapse enough. Against the morphology reference (central brain only), FAFB → BANC reaches 32.3 % after NTAC; the optic lobe reaches 67.4 % against BANC's own types.

Density. BANC's brain graph is sparse: the median neuron has 30 input synapses from other neurons in the graph and 4 input partners (FlyWire: 152 and 14). Thinned copies of FlyWire, which keep each synapse with probability p before the threshold of 5, reproduce BANC's statistics at p ≈ 0.35–0.4. Transfer between independently thinned copies isolates density from all other differences (Figure S4).

- **Density explains about half of the drop on BANC**: the full-density GNN reaches 72.9 % on a copy at BANC's density, against 55.7 % on BANC.
- **A sparser target is best served by a source of its density** (83.9 vs. 72.9 % at p = 0.4), but accuracy falls even then (95.1 → 67.7 % on the diagonal): a sparse reconstruction loses information.
- **A denser target is benign with prior matching**: a GNN trained at p = 0.4 reaches 87.0 % on the full-density copy (70.0 % without prior matching).
- **One GNN trained with random thinning approaches the density-matched GNNs on every target** (94.0, 90.9, 83.7, 63.7 % against 95.1, 91.3, 83.9, 67.7 %), without knowing the target's density.
- On BANC itself, neither density-matched nor augmented GNNs beat the full-density GNN (51.2 and 55.0 vs. 55.7 %, exploratory). There, the density-aligned variant, which lowers the threshold to 2 synapses, recovers strong connections that survive as weak ones; the thinned copies cannot show this, because FlyWire's public connections start at 5 synapses.

Selection. Degraded copies of the development connectomes with known labels (FAFB → MCNS at 100, 50 and 20 % of MCNS's synapses, MCNS → FAFB at 100 and 35 %, FAFB → MCNS without R1-6, with and without bounded prior matching) reproduce the BANC failure: at 20 % density, type consistency ranks the candidates inversely to their accuracy (Spearman −0.68 for cell types). Table S4 compares label-free selection rules.

Table S4: Gap between the chosen and the best candidate preprocessing, points, mean / worst.

Rule	Sandbox (14 cases)	BANC re-scored (6 cases, exploratory)
Type consistency (original)	4.1 / 26.7	19.9 / 56.6
Type sizes closest to the prior	4.1 / 26.0	0.6 / 1.8
Consistency + seed agreement, collapse guard	1.1 / 8.9	1.5 / 5.8
Type sizes + seed agreement (size-and-agreement rule)	**1.3 / 4.0**	**1.1 / 6.3**

Cases are runs × labels. BANC: the four pre-registered runs (FAFB → BANC and MCNS → BANC, cell types and super classes) and, exploratory, FAFB → BANC with GNNs trained on FlyWire thinned to p = 0.4 (both labels). The worst case of the size-and-agreement rule (6.3) is the density-matched super-class run.

Table S5: Hemibrain, accuracy % / macro-F1, as pre-registered (size-and-agreement rule, coverage prior).

Label	Without prior matching	With prior matching	+ NTAC	Best candidate (post hoc)	Type-consistency rule
Cell types	17.3 / 0.09	19.1 / 0.09	**19.6 / 0.09**	25.5	7.8
Super classes	72.8 / 0.20	74.3 / 0.20	**74.4 / 0.19**	78.4	72.8

Type consistency is the best single signal on complete graphs and fails on sparse ones; the type-size divergence behaves the other way round; seed agreement tracks accuracy closely on complete graphs (Spearman 0.98 for MCNS → FAFB cell types). The size-and-agreement rule had the smallest worst case on the sandbox and was chosen there, before it was computed on BANC. Re-scored on BANC (the candidates' accuracies were known from the pre-registered runs), it lands within 0.5 points of the best candidate in all four pre-registered runs, including the collapsed one (58.0 instead of 1.4 %). In the two pre-registered rare-type tests on new targets it chose the best candidate in four of five stage-1 runs and missed by 1.9 points in the fifth. Two candidate fixes did not help: bounding prior matching to avoid filling missing types (fewer wrong R1-6, but lower accuracy), and estimating target type sizes by EM (Saerens et al., 2002), which amplified the classifier's errors.

Partial volumes. The hemibrain covers the central brain without the optic lobes; 77 % of FlyWire's labeled neurons belong to types it does not contain. This breaks the shared assumption of prior matching, the size-based selection and normalization by the target's own statistics: that the target has the source's composition. In the sandbox (MCNS restricted to the central brain, 16 % cropped), the best candidate for the same central-brain neurons fell from 69.9 % (full volume) to 46.9 % (partial) and 36.3 % (partial and cropped); the coverage prior raised it from 30.5 to 36.3 % and was adopted. The second pre-registered test (`docs/hemibrain_protocol.md`, `161cf7b`) was also the first test of the size-and-agreement rule. The harmonized hemibrain edgelist repeats rows (931,329 exact copies), which we drop; within its volume the hemibrain then has FlyWire's density (Table S5).

Of four expectations, the selection was within 5 points of the best candidate for super classes (4.1) but not for cell types (6.4); the coverage prior gave a better best candidate than the FlyWire prior, by little for cell types (25.5 vs. 24.9 %) and decisively after selection for super classes (74.4 vs. 34.5 %); complete neurons were typed better than cropped ones (22.8 vs. 12.2 % on cell types); and cell-type accuracy stayed below 60 %. Super-class accuracy is barely above predicting the most frequent class (73.1 %); visual projection neurons, whose dendrites lie in the missing optic lobe, are typed correctly in 5 % of cases, central-brain neurons in 28 %. Partners outside the volume and cut neurons remove the context the model relies on.

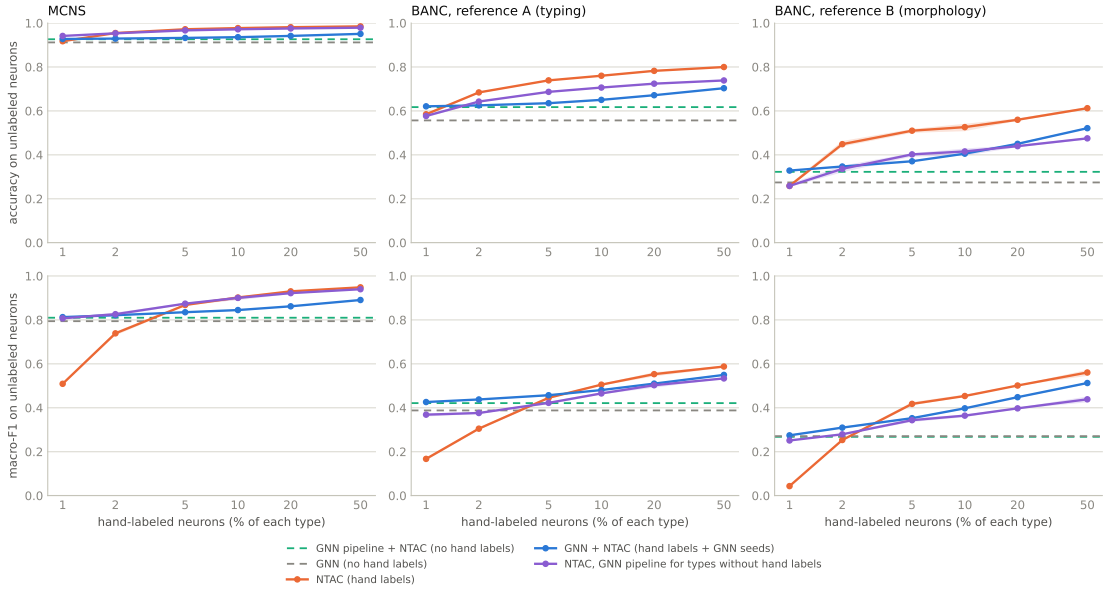


Figure S5: Accuracy and macro-F1 on the neurons without a hand label, against the share of each type labeled by hand (mean over three draws, ≥ 20 label set). Dashed: the label-free pipeline without hand labels.

Table S6: Hand labels NTAC alone needs (% of each type) to match GNN-seeded NTAC at k = 1 %.

Target	GNN + NTAC at 1 %	NTAC alone, by accuracy	NTAC alone, by macro-F1
MCNS	92.7 / 0.81	1.2 %	3.4 %
BANC, BANC's types	62.1 / 0.43	1.3 %	4.4 %
BANC, morphology reference	32.8 / 0.27	1.3 %	2.3 %

S5 Stage 1 next to hand labels, and estimating accuracy without labels

A team typing a new connectome would not rely on wiring alone; it can label neurons by hand and propagate the labels with NTAC. On the ≥ 20 label set, we labeled k % of each type's neurons at random (three draws) and compared NTAC seeded with the hand labels only, NTAC seeded with the hand labels plus the GNN predictions on which three seeds agree (fixed seeds, as in Supplement S2), and NTAC with the label-free pipeline as a fallback for types that have no hand label (Figure S5, Table S6).

By accuracy, the GNN is worth about 1.3 % hand labels on all three references; from 2 % on, NTAC alone is better (MCNS at 5 %: 97.2 vs. 93.2 %; BANC: 73.9 vs. 63.5 %). The GNN's contribution is larger for small types: random labeling at 1 % leaves types with fewer than 50 neurons unlabeled, which NTAC cannot assign, and GNN-seeded NTAC matches the macro-F1 that NTAC alone reaches only with 2.3–4.4 times as many hand labels. Fixed GNN seeds are the wrong way to combine both once hand labels exist, because NTAC never revises them: in the MCNS central brain, GNN-seeded NTAC gains 9 points between 1 and 50 % hand labels, NTAC alone 27 points. The fallback variant is best at 1 % on MCNS (94.1 %, macro-F1 0.81) but hurts on BANC, where the transferred labels are less reliable. The rare-type pipeline was not part of this comparison.

Can one tell without labels how good a labeling is? We fitted simple linear predictors of accuracy from the label-free properties of all 1,200 candidate labelings of the development runs

and the sandbox, and tested them on BANC and the hemibrain. The pre-specified predictor (seed agreement and type-size divergence) missed by 12.8 points on average (median 8.4). It was accurate on complete targets with a non-collapsed labeling (FAFB → BANC cell types: 0.5 points). It failed when all seeds made the same error: on the collapsed MCNS → BANC labeling (predicted 44.5 %, true 1.4 %) and on hemibrain super classes with the FlyWire prior (seed agreement 0.94, 34.7 % correct). Seed agreement does rank neurons and types where the transfer works: reviewing the 20 % least-agreed neurons finds 77 % of the errors on MCNS, 55 % on FlyWire and 34 % on BANC (random review: 20 %), and per type it correlates with F1 (Spearman 0.54–0.66). On the hemibrain, where 80 % of the labels are wrong, it is no better than random.

S6 Macro-F1 of the rare-type results

Table S7: Macro-F1 (mean F1 over the true types present) for the rare-type tables of the main text; accuracies are given there. –: not run, or not computed (the GNN-free combination leaves some neurons without a type).

Table 3, development	FlyWire left → right	FlyWire → MCNS
Stage 1	0.78	0.49
Stage-1 probabilities, one-to-one	–	0.62
Two-stage, strict candidates	0.88	0.78
Two-stage, broad candidates	–	0.83
GNN votes, one-to-one (oracle candidates)	0.81	–
Graph matching, ipsilateral (oracle candidates)	0.92	–
Graph matching (oracle candidates)	0.93	0.87

Table 4, pre-registered tests	BANC brain	BANC cord	MCNS cord
Stage 1	0.19	0.36	0.70
Two-stage, strict candidates	0.47	0.55	0.79
Two-stage, broad candidates	0.57	0.61	0.82
Oracle candidates	0.70	0.73	0.86

Table 5, candidate selection	MCNS thinned	MCNS full
Strict	0.53	0.78
Broad	0.65	0.83
Oracle	0.84	0.87

Table 6, wiring and morphology	BANC brain	BANC cord
NBLAST alone	0.65	0.29
Two-stage, broad candidates	0.57	0.61
Two-stage, morphology-augmented candidates	0.67	0.63
Combination, broad candidates	0.68	0.59
Combination, morphology-augmented candidates	0.71	0.61
Combination, NBLAST-only candidates (no GNN)	–	–
Two-stage, seeded	0.65	0.53
Combination, seeded (pre-specified)	0.69	0.51